

An Introduction to Measure Theory - Chapter 1

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1 Exercise 1.1.1 - Boolean Closure

Show that if $E, F \subset \mathbb{R}^d$ are elementary sets, then the union $E \cup F$, the intersection $E \cap F$, and the set theoretic difference $E \setminus F := \{x \in E : x \notin F\}$, and the symmetric difference $E \Delta F := (E \setminus F) \cup (F \setminus E)$ are also elementary. If $x \in \mathbb{R}^d$, show that the translate $E + x := \{y + x : y \in E\}$ is also an elementary set.

An **elementary set** is that which is a union of a finite amount of boxes. Thus, let $\bigcup_i E_i$ and $\bigcup_j F_j$ be the unions of the finite number of boxes comprising sets E and F , respectively.

$$E \cup F = \bigcup_i E_i \cup \bigcup_j F_j = \bigcup_i \bigcup_j E_i \cup F_j \quad (1)$$

$$E \cap F = \bigcup_i E_i \cap \bigcup_j F_j = \bigcup_i \bigcup_j E_i \cap F_j \quad (2)$$

$$E \setminus F = \bigcup_i E_i \setminus \bigcup_j F_j = \bigcup_i \left(E_i \cap \neg \bigcup_j F_j \right) = \bigcup_i \left(E_i \setminus \bigcup_j F_j \right) \quad (3)$$

This one tripped me up a bit because elementary sets aren't closed under set negation, and I was inherently proving the closure of elementary sets under the set difference for some reason.

$$E \Delta F = E \setminus F \cup F \setminus E = \bigcup_i \left(E_i \setminus \bigcup_k F_k \right) \cup \bigcup_j \left(F_j \setminus \bigcup_k E_k \right) = \bigcup_i \bigcup_j \left(E_i \setminus \bigcup_k F_k \cup F_j \setminus \bigcup_k E_k \right) \quad (4)$$

$$E + x = \bigcup_i (E_i + x) \quad (5)$$

The set theoretical operations of union, intersection, difference, and symmetric difference, and translation, between two elementary sets yield a union of finitely countable boxes, which is itself an elementary set. $\square$

2 Exercise 1.1.2

Give an alternate proof of Lemma 1.1.2(ii) by showing that any two partitions of E into boxes admit a mutual refinement into boxes that arise from taking Cartesian products of elements from finite collections of disjoint intervals.

Lemma 1.1.1(ii) If E is partitioned as the finite union $B_1 \cup \dots \cup B_k$ of disjoint boxes, then the quantity $m(E) := |B_1| + \dots + |B_k|$ is independent of the partition. In other words, given any other partition $B'_1 \cup \dots \cup B'_{k'}$ of E , one has $|B_1| + \dots + |B_k| = |B'_1| + \dots + |B'_{k'}|$.

$$\begin{aligned} |E| &= \left| \bigcup_k B_k \right| = \left| \bigcup_{\tilde{k}} B_{\tilde{k}} \right| \\ &= \left| \bigcup_k I_{k,1} \times \dots \times I_{k,d} \right| = \left| \bigcup_{\tilde{k}} I_{\tilde{k},1} \times \dots \times I_{\tilde{k},d} \right| \end{aligned} \quad (6)$$

Let $\{J_{m,1}, \dots, J_{m,d}\}$ be the disjoint intervals between all $2k$ endpoints of $\{I_{k,1}, \dots, I_{k,d}\}$ and $\{\tilde{J}_{\tilde{m},1}, \dots, \tilde{J}_{\tilde{m},d}\}$ be the disjoint intervals between all $2\tilde{k}$ endpoints of $\{I_{\tilde{k},1}, \dots, I_{\tilde{k},d}\}$. Let $l_{k,i}$ for $i \in [d]$ denote the disjoint intervals from $J_{m,i}$ that union to form $I_{k,i}$ and $\tilde{l}_{\tilde{k},i}$ denote the disjoint intervals from $\tilde{J}_{\tilde{m},i}$ that union to form $I_{\tilde{k},i}$.

$$|E| = \left| \bigcup_k \left(\bigcup_{l_{k,1}} J_{l_{k,1}} \times \dots \times \bigcup_{l_{k,d}} J_{l_{k,d}} \right) \right| = \left| \bigcup_{\tilde{k}} \left(\bigcup_{\tilde{l}_{\tilde{k},1}} J_{\tilde{l}_{\tilde{k},1}} \times \dots \times \bigcup_{\tilde{l}_{\tilde{k},d}} J_{\tilde{l}_{\tilde{k},d}} \right) \right| \quad (7)$$

Thus, it has been shown that any two partitions of E into boxes admit a mutual refinement into boxes that arise from taking the Cartesian product from elements from finite collections of disjoint intervals. $\square$

3 Exercise 1.1.2.5

Give some examples of sets for which the limit formula $m(e) = \frac{1}{N} \#(E \cap \frac{1}{N} \mathcal{N}^d)$ does not exist.

Example 1. The limit can fail if it diverges to $\pm\infty$. This occurs when the interval is unbounded.

$$E = \mathbb{R}^d \quad (8)$$

Example 2. Even if bounded, the limit can fail if it oscillates.

$$E = \bigcup_{k=1}^{\infty} \left(\frac{1}{2^{2k}}, \frac{1}{2^{2k-1}} \right) \quad (9)$$

This is the set of alternating dyadic intervals close to 0.

4 Exercise 1.1.3 - Uniqueness of Elementary Measure

Let $d \geq 1$. Let $m' : \mathcal{E}(\mathbb{R}^d) \rightarrow \mathbb{R}^+$ be a map from the collection $\mathcal{E}(\mathbb{R}^d)$ of elementary subsets of $\mathbb{R}^d$ to the nonnegative reals that obeys the non-negativity, finite additivity, and translation invariance properties. Show that there exists a constant $c \in \mathbb{R}^+$ such that $m'(E) = cm(E)$ for all elementary sets E . In particular, if we impose the additional normalisation $m'[0, 1)^d = 1$ then $m' \equiv m$. Hint: Set $c := m'([0, 1)^d)$, and then compute $m'([0, \frac{1}{n})^d)$ for any positive integer n .

The half-open sets in the hint indicate that finite additivity corresponding to tiling is involved in this proof. Further, the $\frac{1}{n}$ in the hint indicates that the set to be measured is broken into n^d tiles.

Let $c = m'([0, 1)^d)$ be a scalar corresponding to the m' measure of the unit box. Partition the unit box into n^d tiles as: $[0, 1)^d = \bigcup_{i=1}^{n^d} ([0, 1/n)^d + x_i)$. Then,

$$\begin{aligned} m'([0, 1)^d) &= m' \left(\bigcup_{i=1}^{n^d} ([0, 1/n)^d + x_i) \right) \\ &= \sum_{i=1}^{n^d} m'([0, 1/n)^d + x_i) && \text{By finite additivity.} \\ &= \sum_{i=1}^{n^d} m'([0, 1/n)^d) && \text{By translational invariance.} \\ &= n^d m'([0, 1/n)^d) \end{aligned} \tag{10}$$

Similarly, $m([0, 1)^d) = n^d m([0, 1/n)^d)$. Rewrite c as the ratio of these two measures:

$$\begin{aligned} c &= \frac{n^d m'([0, 1/n)^d)}{n^d m([0, 1/n)^d)} \\ &= \frac{m'([0, 1/n)^d)}{m([0, 1/n)^d)} \\ &= \frac{m'([0, 1)^d)}{m([0, 1)^d)} \end{aligned} \tag{11}$$

Thus, by finite additivity of the elementary set's decomposition into finite disjoint unions of half-open boxes, the existence of a unique c for all $E \in \mathcal{E}^d$ has been proved. $\square$

4.0.1 Scratch Work

From non-negativity and finite additivity, monotonicity is derived: Let X be a subset of Y . Then, there exists some elementary set $Z = Y \setminus X$ disjoint to X such that $Y = X \cup Z$.

$$\begin{aligned} m(Y) &= m(X) + m(Z) \quad \forall X, Z : X \cap Z = \emptyset && \text{By finite additivity.} \\ m(X) &= m(Y) - m(Z) \\ m(X) &\leq m(Y) && \text{By non-negativity of } Z. \end{aligned} \tag{12}$$

From this, finite subadditivity may be derived:

$$\begin{aligned}
m(X \cup Y) &= m(X \cup (Y \setminus X)) \\
&= m(X) + m(Y \setminus X) && \text{By finite additivity.} \\
&\leq m(X) + m(Y) && \text{By monotonicity of } m(Y \setminus X) \text{ and } m(Y).
\end{aligned} \tag{13}$$

Now, let E be the union of a finite amount of elementary sets, $E = \bigcup_i E_i$. By finite subadditivity, $m(E) \leq \sum_i m(E_i)$, and by scalar multiplication, $c * m(E) \leq c * \sum_i m(E_i)$

Let $c = m'([0, 1]^d)$.

$$\begin{aligned}
cm(E) &= m'([0, 1]^d)m(E) \\
&\leq m'([0, 1]^d) \sum_i m(E_i) \\
&= \sum_i m'([0, 1]^d)m(E_i)
\end{aligned} \tag{14}$$

$$m'([0, \frac{1}{n}]^d) = m'([0, 1]^d)m([0, \frac{1}{n}]^d) \tag{15}$$